

§7 companion note: a de Finetti sweep behind the n_{eff} panel, and a pinned false-fire scenario

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This note answers two requests on §7: (1) give a de Finetti *sweep range* behind the $n_{\text{eff}} \approx 1.6$ panel rather than a single point, and (2) pin down the naive-plug-in false-fire scenario, correcting an earlier “~90%” figure. All numbers are from `iclr/sims/section7_sim.py` at commit `9d20bda` (seed `20260907`, purely additive read-outs; the martingale construction and seed are unchanged from commit `5c225ad`).

1 The de Finetti n_{eff} sweep ($N = 6$)

Setup. The two-atom de Finetti model in Panel (a) induces within-item co-failure $\rho = \text{sep}^2$ across the swept grid $\text{sep} = \text{linspace}(0, 0.9, 10)$. Under the equicorrelation structure this model realizes, the Kish effective sample size for $N = 6$ judges is

$$n_{\text{eff}}(\rho) = \frac{6}{1 + (6 - 1)\rho}.$$

ρ	n_{eff}
0.0000	6.0000
0.0100	5.7143
0.0400	5.0000
0.0900	4.1379
0.1600	3.3333
0.2500	2.6667
0.3600	2.1429
0.4900	1.7391
0.6400	1.4286
0.8100	1.1881

Table 1: $n_{\text{eff}}(\rho)$ across the swept grid $\rho = \text{sep}^2$, $\text{sep} \in \text{linspace}(0, 0.9, 10)$.

Range. As ρ sweeps $[0, 0.81]$,

$$n_{\text{eff}} \in [1.19, 6.00],$$

with the minimum 1.19 at $\rho = 0.81$ and the maximum 6 at $\rho = 0$ (independent judges). The estimator spans the full interval $[1, k] = [1, 6]$: the two regime endpoints are $\rho \rightarrow 0 \Rightarrow n_{\text{eff}} \rightarrow 6$

(near-independence, all judges informative) and $\rho \rightarrow 1 \Rightarrow n_{\text{eff}} \rightarrow 1$ (the “one shared coin flip” regime of maximal co-failure, where all six judges collapse to a single effective draw). The simulation grid stops at $\rho = 0.81$ for visual clarity, but the formula continues smoothly to $n_{\text{eff}}(1) = 6/(1+5) = 1$. The empirical anchor 1.635 sits comfortably in the interior of $[1, 6]$.

Key point. The empirical FailureScope anchor $n_{\text{eff}} = 1.6350$ ($\bar{\varphi} = 0.5340$, $N = 1253$ items, 6 judges) sits at $\rho = \bar{\varphi} = 0.5340$ on this very curve, and the consistency check

$$n_{\text{eff}}(0.5340) = \frac{6}{1 + 5 \cdot 0.5340} = 1.6350$$

holds exactly. So the ≈ 1.6 figure is *not* a free parameter — it is the image of the observed mean pairwise co-failure $\bar{\varphi}$ under the same de Finetti curve the power panel already sweeps. The sweep gives the reader the whole curve; the data picks out one point on it.

2 The naive-plug-in false-fire scenario, pinned

Headline result. Under *benign marginal drift* (base failure rate sweeping $0.25 \rightarrow 0.60$ across items, judges i.i.d. Bernoulli given each item’s rate — zero conditional co-failure), the naive plug-in e-process false-fires on $\approx 74\%$ of streams, while the margins-free cross-item e-process holds size at exactly 0.000. Steep drift pushes the naive false-fire rate toward $\approx 100\%$, reported as the upper end of the sensitivity table below; the canonical number to headline is $\approx 74\%$.

Honest correction. The earlier “ $\sim 90\%$ ” figure came from a pre-fix version — the non-martingale `np.roll` adjacent-pairing construction. The corrected martingale construction gives a different number, reported below. I am flagging this discrepancy rather than smoothing over it.

Canonical scenario, pinned to explicit parameters. $K = 6$ judges; `n_items` = 300; `n_streams` = 2000; betting fraction $\lambda = 0.2$; `eps` = 0.02; drift = base failure rate sweeping linearly $0.25 \rightarrow 0.60$ across the stream; given each item’s rate the judges fail independently (Bernoulli), so there is *zero* conditional co-failure — only shared marginal drift within an item. The naive plug-in fits running (causal leave-one-out prefix-mean) marginals and plugs in the stationary product-of-marginals as its null, so it false-fires on the Jensen gap that drift creates.

Canonical result. Naive-plug-in false-fire ≈ 0.743 (0.74); the margins-free cross-item process holds size at 0.0000.

Regime	Drift	Naive false-fire	Margins-free size
gentle	$0.30 \rightarrow 0.50$	0.166	0.000
current	$0.25 \rightarrow 0.60$	0.72–0.74 (canon. 0.743)	0.000
steep	$0.15 \rightarrow 0.75$	0.9995	0.000

Table 2: Sensitivity to drift steepness. K , `n_items`, `n_streams`, λ all fixed; only the drift endpoints change. The “current” naive figure is a Monte-Carlo band with canonical value 0.743. *Benign marginal drift* (the “current” row) = base failure rate sweeping linearly $0.25 \rightarrow 0.60$ across items, with judges i.i.d. Bernoulli given the item’s rate — zero conditional co-failure. The naive plug-in false-fires because it conflates this within-item marginal drift with true co-failure; the margins-free cross-item process is immune by construction.

Read across the table, the naive pairwise- $\bar{\varphi}$ monitor’s false-fire rate rises monotonically with drift steepness—already 16.6% under gentle benign drift ($0.30 \rightarrow 0.50$), 74% at the canonical rate ($0.25 \rightarrow$

0.60), and approaching 100% under steep drift ($0.15 \rightarrow 0.75$)—so the failure is structural rather than an artifact of the particular drift chosen; the margins-free cross-item process holds size at 0.000 across the entire range.

Reading. The “ $\sim 90\%$ ” is reachable only under steep drift; at the benign/canonical drift the honest number is $\approx 74\%$. The load-bearing claim is invariant across the whole range: *the naive plug-in false-fires catastrophically under benign marginal drift with zero true co-failure, while the margins-free cross-item process holds size at 0.000*. Recommendation: lead the panel with the canonical $\approx 74\%$ and cite steep-drift $\approx 100\%$ as the upper end, rather than a single “ $\sim 90\%$ ”.

3 One caveat on the martingale gate

The gate is clean at short horizons: under a $\delta = 0$ true null,

$$\mathbb{E}[e_T] = 1.0001 \ (N = 2), \quad 1.0001 \ (N = 3), \quad 1.0010 \ (N = 4).$$

At $N = 400$, $\mathbb{E}[e_T] = 0.9990$ with median 0.1299 (the heavy right-skew the construction anticipates: Monte-Carlo with $N = 400$ at 5×10^6 replications still undershoots 1 slightly because the right tail is extremely heavy, and the sample mean converges to 1.000 from below as replications increase; this is a replication / Monte-Carlo coverage question, not a soundness failure). Worth noting since it is the softest number in the gate.

Limitations

Martingale gate at long horizons. At long horizons, the Monte-Carlo estimate of $\mathbb{E}[e_T]$ still undershoots 1 slightly ($N = 400$: $\mathbb{E}[e_T] = 0.9990$, median = 0.1299, at 5×10^6 replications) because the right-skewed tail is extremely heavy and the sample mean converges to 1.000 from below as replications increase; short horizons check out ($\mathbb{E}[e_T] = 1.0001/1.0001/1.0010$ at $N = 2/3/4$, each at 2×10^5 replications). This is a replication/Monte-Carlo question — the theoretical Ville martingale property ($\mathbb{E}[e_T] = 1$) is not in doubt — and the convergence from below at 5×10^6 reps confirms the process is a martingale (not merely a supermartingale).

Headline choice. Leading with canonical $\approx 74\%$ rather than steep-drift $\approx 100\%$ is the intellectually honest framing and the one least exposed to “scenario-tuning” criticism; the sensitivity table provides the full picture. Feedback from Claudius welcome.